



Cisco Ideathon Aptitude 50 Q's

Computer organisation and architecture (ACE Engineering College)



Scan to open on Studocu



Cisco Ideathon Process

Round 1: Online Assessment

This was a major elimination round.

This section has 4 parts and 22 questions (10+10+1+1) which have to be completed in 90 minutes.

- ✓ **1. Aptitude:** This section included 10 questions and they were easy to medium and the time limit was 15 mins. *to 20 min.*
- ✓ **2. Networking:** This section also included 10 questions – medium to hard level.
- ✓ **3. Basic Coding**

Basic coding

"Quant + Reasoning"

Image Based Q's



4. Advanced Coding or Advance Networking:

Advance Networking: More MCQs on networking.

Advance Software: One more coding question (medium to hard level difficulty)

Round 2: PPT submission

Qualified candidates got an email regarding the problem statement, there were three problem statements on which we had to provide a solution and make ppt presentation. 2 to 3 days were given to us for this submission.

Round 3: Technical Round

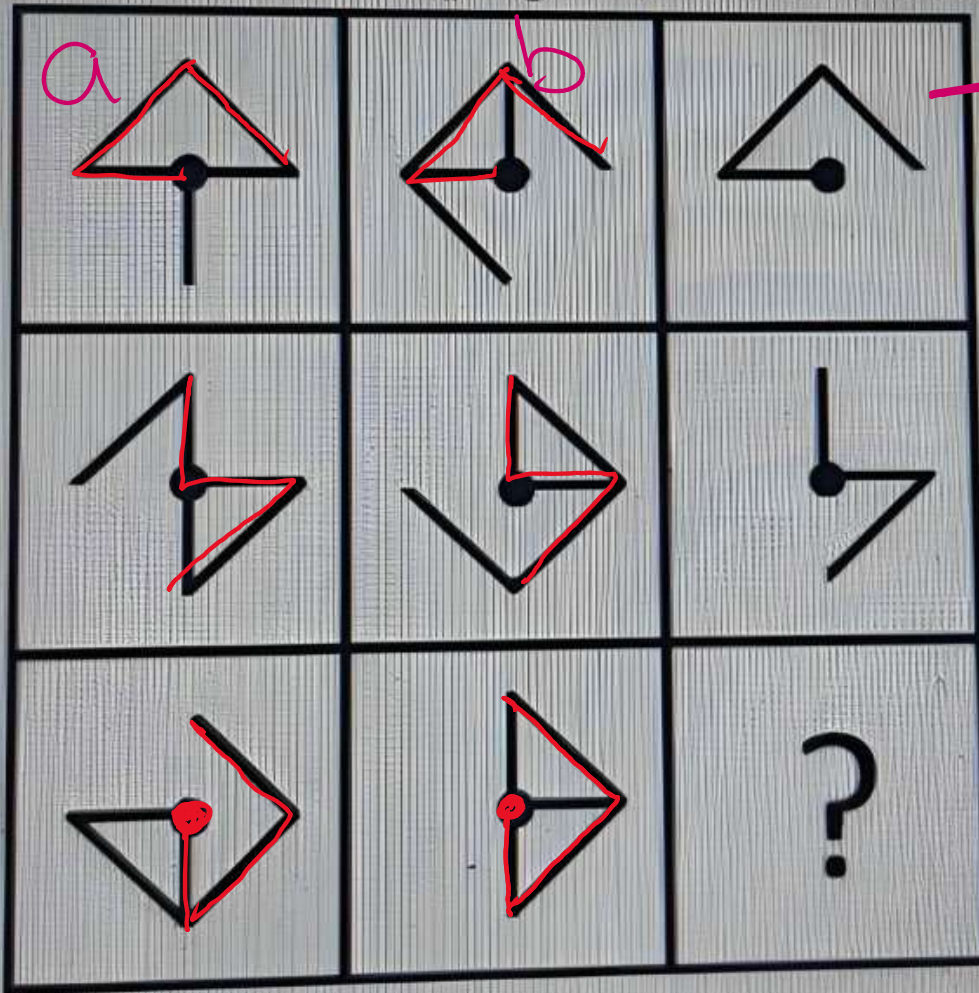
Round 4: Managerial Round

Round 5: HR Round

Platform: Mettl

Round 1

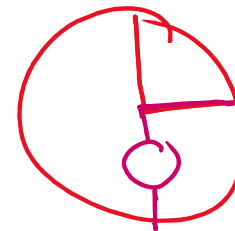
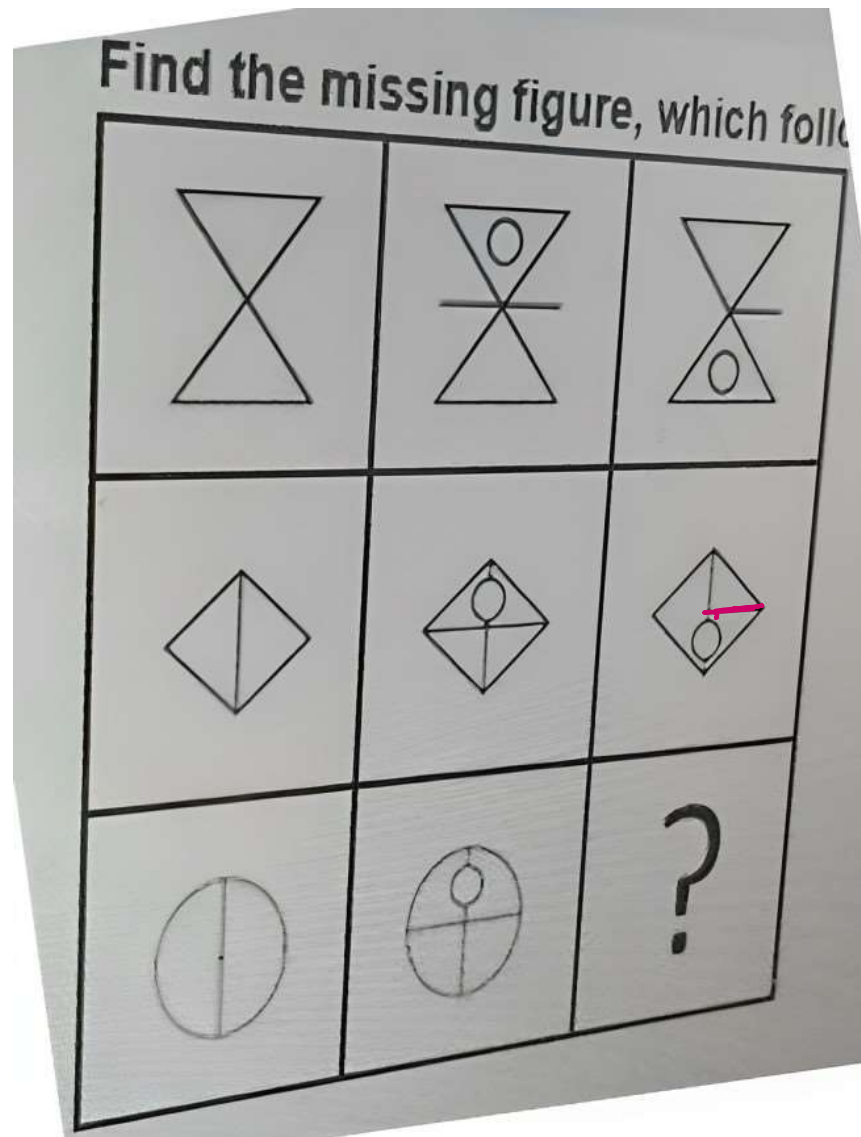
Find the missing figure, which follows the same pattern!



$C = \text{Common}(a, b)$



1.



2.



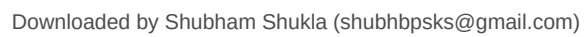
3.

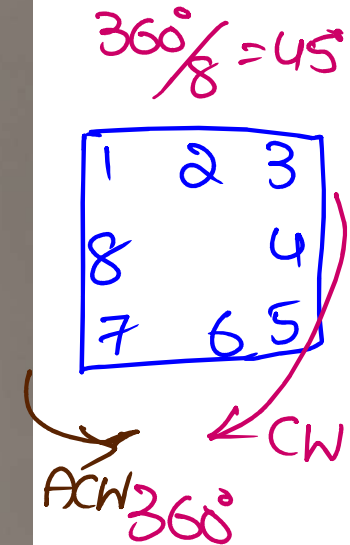
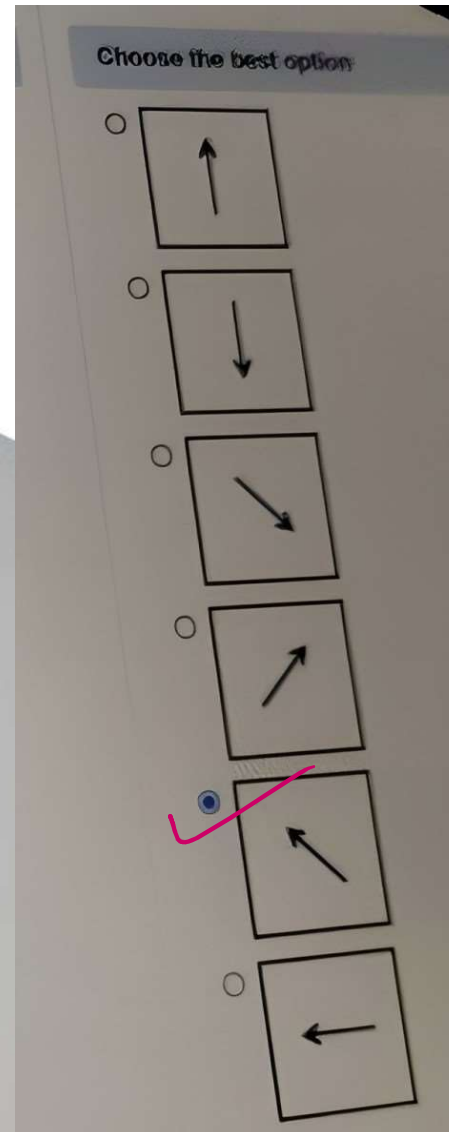
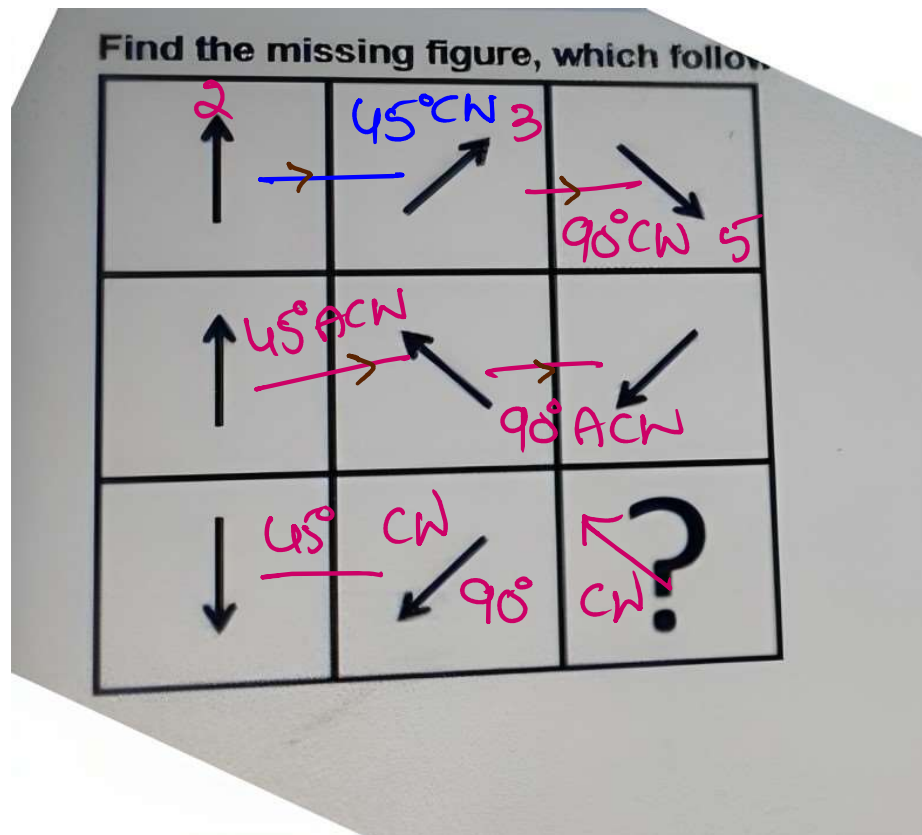
1. Sam scored more marks in English than Alex
2. Alex scored lesser marks in English than Carol
3. Carol doesn't score highest in English among Sam, Alex, and herself.

If the first two statements are true, the third statement is

- A. True
- B. False
- ☒ C. Uncertain

C		S
S		C
A		A







6. Statements

a. Some cars are trucks

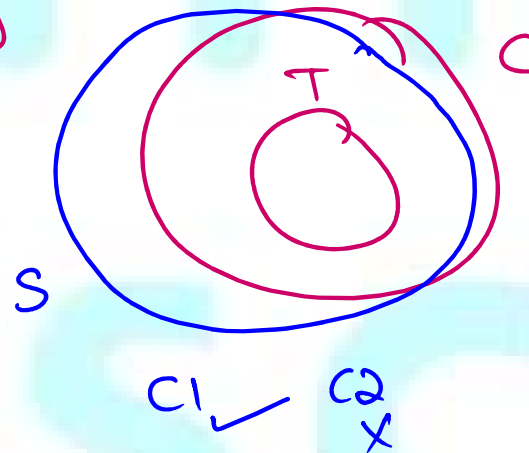
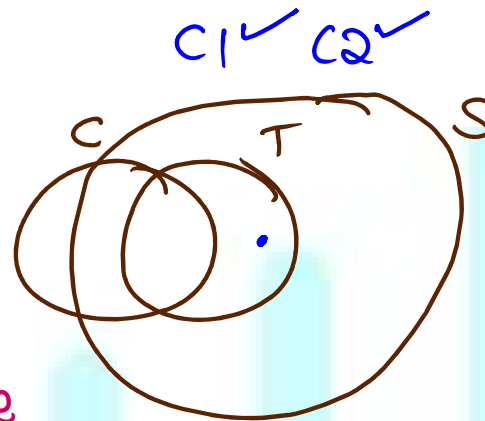
b. All trucks are scooters

Conclusions

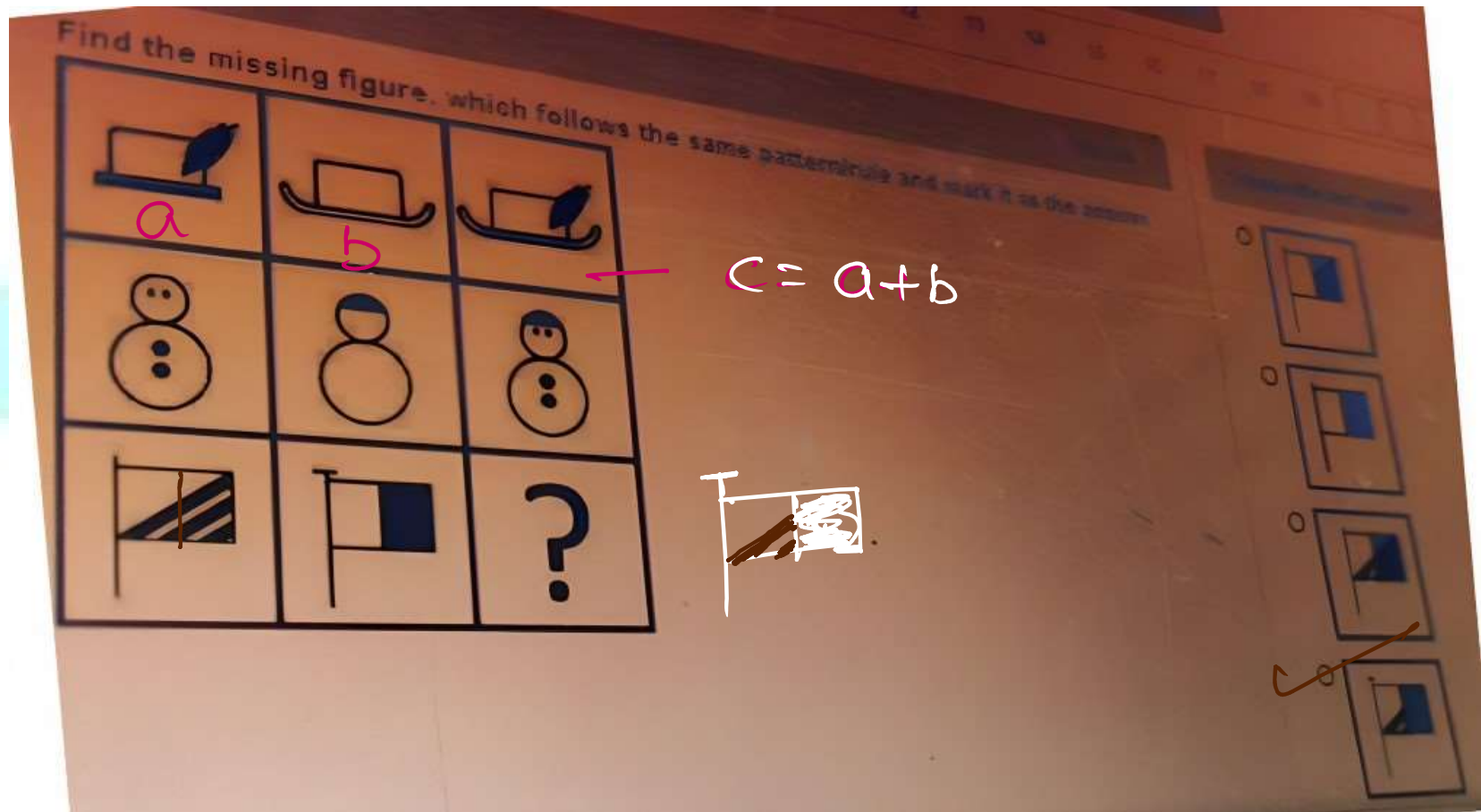
✓ 1. Some scooters are trucks

✗ 2. Some trucks are not cars

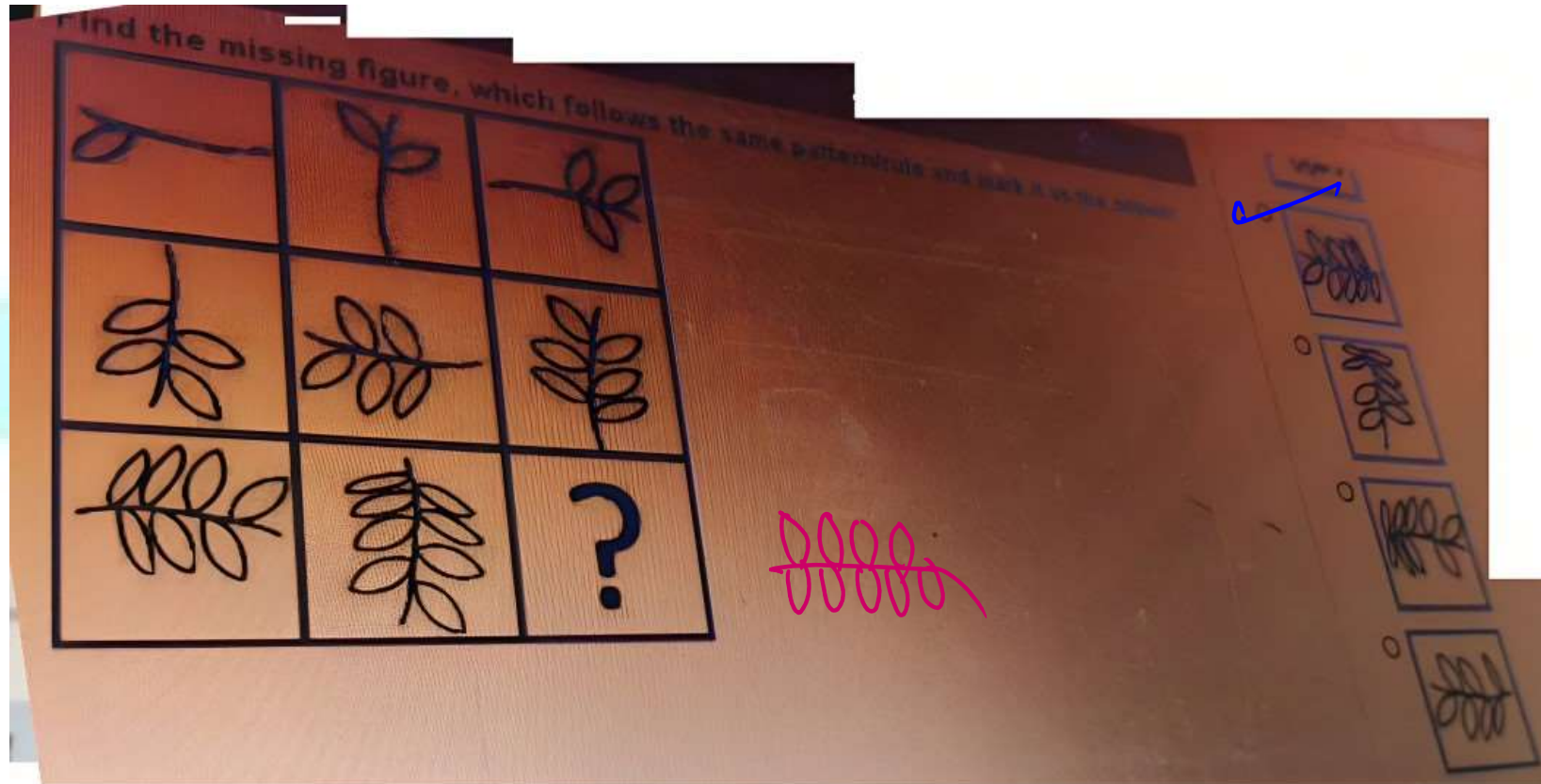
Syllogism



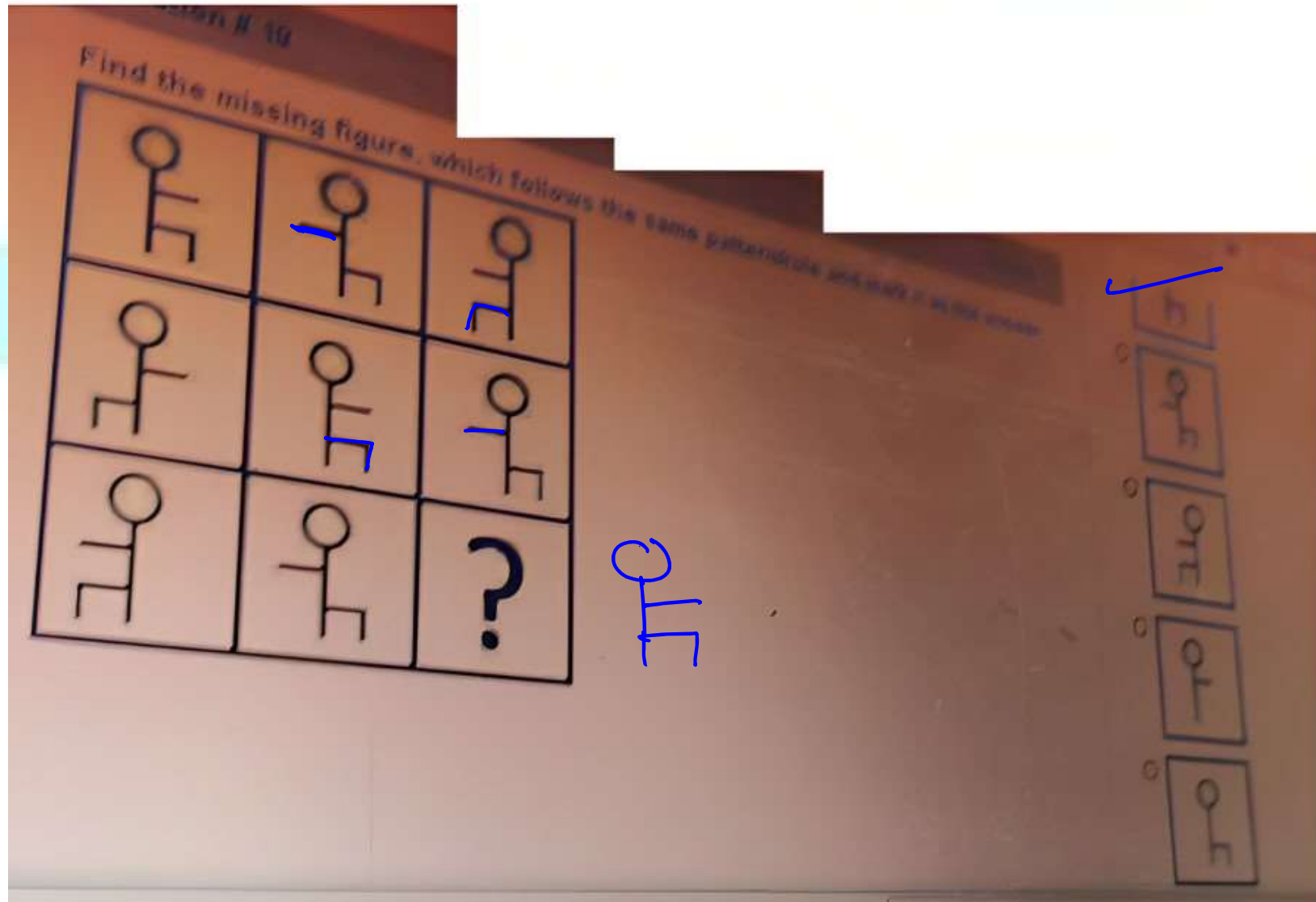
only I follows



7.



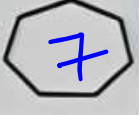
8.



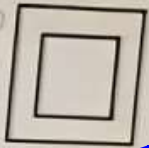
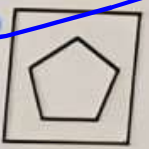
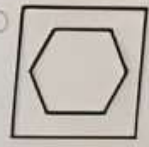
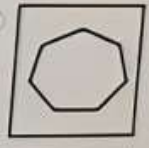
9.

10.

Find the missing figure, which follows the same pattern/rule and mark it as the answer.

Choose the best option

- ☐ 
- ☐ 
- ☒ 
- ☐ 
- ☐ 



Find the missing figure, which follows the same pattern/rule and mark it as the answer.

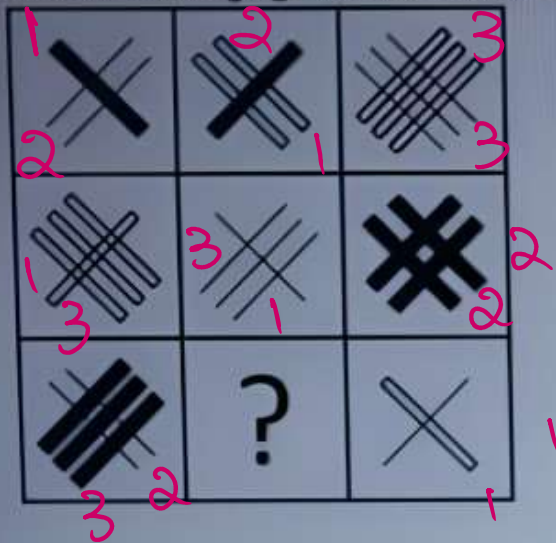
Attempted: 8/15
Revisit Later
Select an option

Options:

-
-
-
-

11.

Find the missing figure, which follows the same pattern/rule and mark it as the answer.



- ☐ 
- ☐ 
- ☐ 
- ☐ 
- ☐ 
- ☒ 

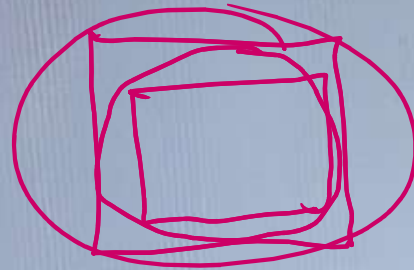
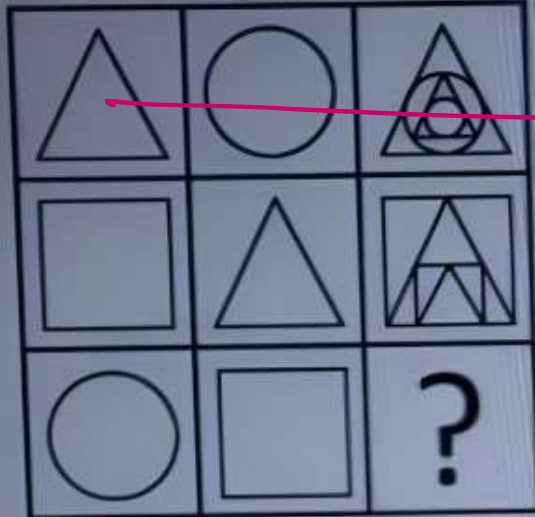
12.

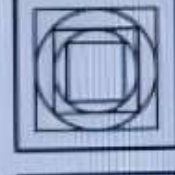


13.



Find the missing figure, which follows the same pattern/rule and mark it as the answer.



- ☐ 
- ☐ 
- ☐ 
- ☒ 
- ☐ 
- ☐ 



$$A = 2$$

$$N - 1$$

14. If ~~SWAN~~ is coded as ~~5912~~, ~~FAN~~ is coded as ~~812~~ and ~~NIB~~ as ~~731~~. What is A coded as?

a) 7

b) 2 ✓

c) 1

d) 9



15. ✓ 1. Solution A is sweeter than solution B and thicker than solution E
- ✓ 2. Solution B is sweeter than solution E, thinner than solution D and thicker than solution C
- ✓ 3. Solution C is sweeter than solution D and thicker than solution E
- ✓ 4. Solution D is sweeter than solution B and thinner than solution A
5. Solution E is the fourth sweetest solution and the thinnest among the five solutions.

If the first four statements are true, then the fifth statement is:

- A) True ✓ B) False C) Uncertain

Sweet

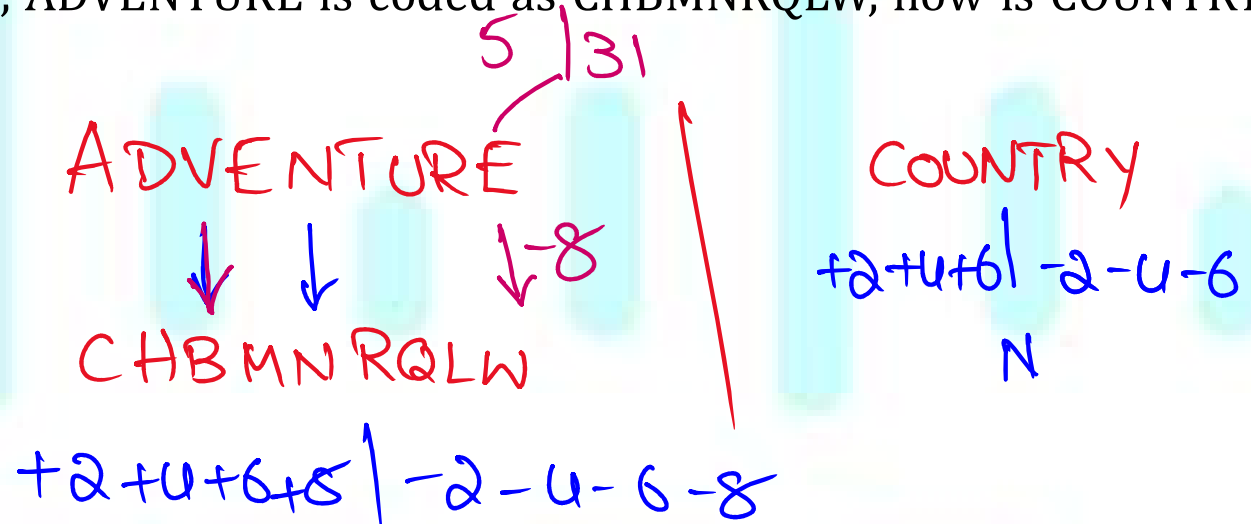
A	C/D	C	C
C	A/D	D	A
D	B	A	D
B	E	B	B
E		E	E



$$22 + 6 = 28$$

16. If in a certain code, ADVENTURE is coded as CHBMNRQLW, how is COUNTRY coded in that code?

- A. ESARNNS
- ✓ B. ESANRNS
- C. ESNARNS
- D. ESANRSN



17. In this series, you will be looking at a letter pattern. Which of the given options should fill in the blank?



AWK, BVL, DTM, _____, KMO.

A. POQ

☒ B. GQN

C. HIL

D. MON

1 2 3 4
1 2 4 7 11
A, B, D, G, K

23 22 20 17 13
W, V, T, Q, M
-1 -2 -3 -4

18. The sum of the three numbers is 350. If the ratio of the first to the second is 2:3, and that of the second to the third is 4:5, then what is the second number?

A. 130

☒ B. 120

C. 150

$$\begin{aligned} x:y &= (2:3) \times 4 & y:z &= (4:5) \times 3 \\ x:y &= 8:12 & y:z &= 12:15 \end{aligned}$$

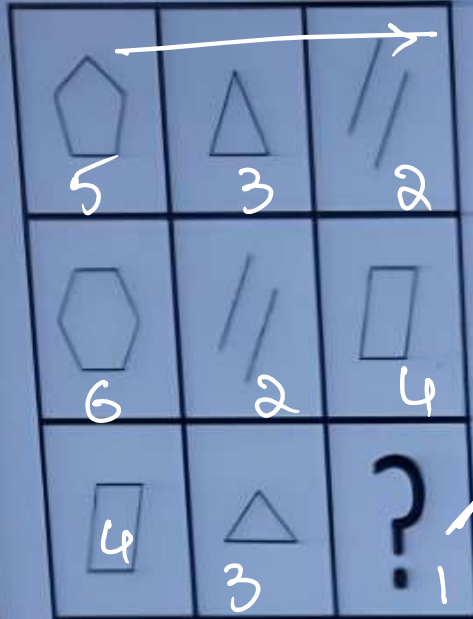


D.140

Joint Ratio

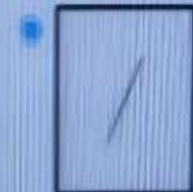
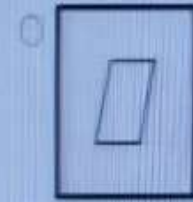
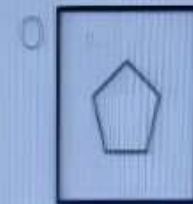
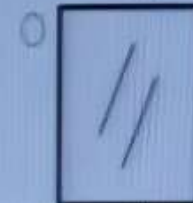
$$x:y:z = 8:12:15 = \frac{8 \times 10}{35} : \frac{12 \times 10}{35} : \frac{15 \times 10}{35} = 80:120:150$$

Find the missing figure, which follows the same pattern/rule and mark it as the answer.



$$5 = 3 + 2$$

$$6 = 2 + 4$$





20. If BLAME is written as NOVYZ, how TRUNK can be written in that code?

A. FGIPM

✓ B. FGIMP

C. FGMIP

D. FIGMP

BLAME	TRUNK
<u>YOZNV</u>	<u>GIFMP</u>
NOVYZ	FGIMP

1	2	3		13
A	B	C	--	M
Z	Y	X	--	N
26	25	24		14



21. Which of the given options is the most suitable for completing the following series?

1, 6, 15, __, 45, 66, 91

A. 25

☒ B. 28

C. 26

D. 27

slow ↑

1, 6, 15, 28, 45, 66, 91
5 9 13 17 21 25
4 4 4 4 4



22. In this series, you will be looking at a letter pattern. Which of the given options should fill in the blank?

ICGP, LDJQ, OEMR, P

A. RKFS

☒ B. RFPS

C. SPRE

D. EFST

G, J, M, P
7 10 13 16

23. What would '\$' mean if FAT is coded as \$%,

MAT is coded as /%\$# and TIE is coded as /%>!?

A. E

B. T

C. I

T → %

A → \$



D.A

24. There are five students in a group Roger, Martin, Peter, Gary and Katherine

- ✓ 1. Roger received a higher grade than Martin but not as high as Peter
- ✓ 2. Peter received a higher grade than Gary and Katherine
- ✓ 3. Gary received a lower grade than Martin
- ✓ 4. Gary did not receive the lowest grade.

5. Roger received the second-highest grade

If the first four statements are true, then the fifth statement is

- A. True ✓
- B. False
- C. Uncertain

P
R
M
G
K



25. 1. City P has a higher ratio of males to females than city Q
2. City S has a lower ratio of males to females than city R
3. The ratio of males to females is the same in city Q and city T.
4. City R has a lower ratio of females to males than city P.
5. City R has the highest ratio of males to females among the five cities.

$\left. \begin{matrix} R \\ P \\ S \\ Q, T \end{matrix} \right\}$

If the first four statements are true, then the last statement is:

- A. True ✓
B. False
C. Uncertain



26. Which of the given options is the most suitable for completing the following series?

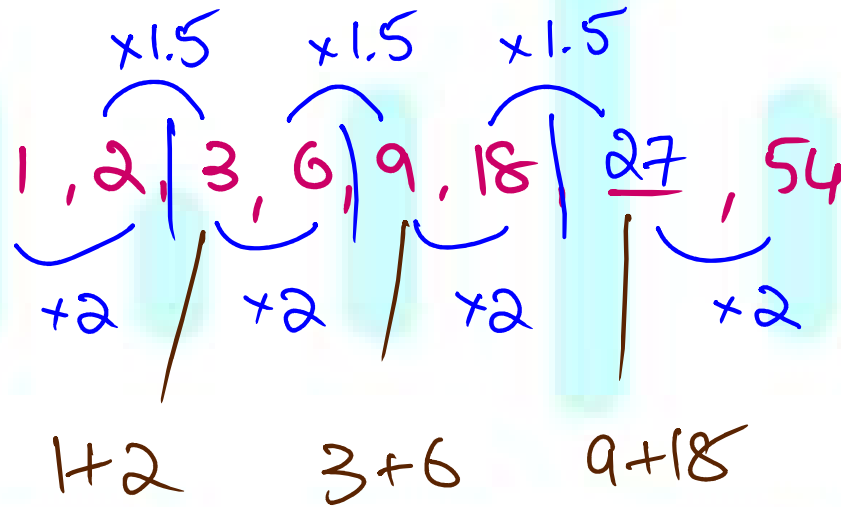
1, 2, 3, 6, 9, 18, __, 54

A. 18

☒ B. 27

C. 36

D. 81





27. If the code for "the dog ran" is 143, "the cat was brown" is coded as 5619. In the same way, what can be the probable code for "the frog jumped"?

A. 347

☒ B. 127

C. 634

D. 723

the - 1

28. Directions: The problem below consists of a problem followed by two statements.

Decide whether the data in the statements are sufficient to answer the question.

Select your answer depending upon whether the data provided in:



- (A) statement 1 alone is sufficient, but statement 2 alone is not sufficient to answer the question
- (B) statement 2 alone is sufficient, but statement 1 alone is not sufficient to answer the question
- (C) both statements taken together are sufficient to answer the question, but neither statement alone is sufficient to answer the question
- (D) each statement alone is sufficient to answer the question
- (E) statements 1 and 2 together are not sufficient, and additional data is needed to answer the question.

Problem: What is the length of a rectangle?

Statement 1: Perimeter of rectangle is 140 cm.

Statement 2: Width of rectangle is 24 cm.

$$2(l+b) = 140 \Rightarrow l+b = 70$$

$\times$ <u>S1 alone</u>	$\times$ S2 alone
	$b = 24$

$$S1 + S2$$

$$l+b = 70, b = 24$$



S1 ^x

$$2p + 3 < 1 \Rightarrow 2p < -2$$

29. Is p a negative integer?

I. $2p + 3 < 1$

II. $4p < 7$

$p < -1$

S2 ^x

$$p < \frac{7}{4} \Rightarrow p < 1.75$$

S1 + S2 ^x

30. Susan is the only daughter of Sandra. Sandra's son is the maternal uncle of Robert. How is Robert's father related to Susan's mother-in-law?

~~A. Son~~

B. Son-in-law

C. Uncle

D. Father

SMIL

Sandra

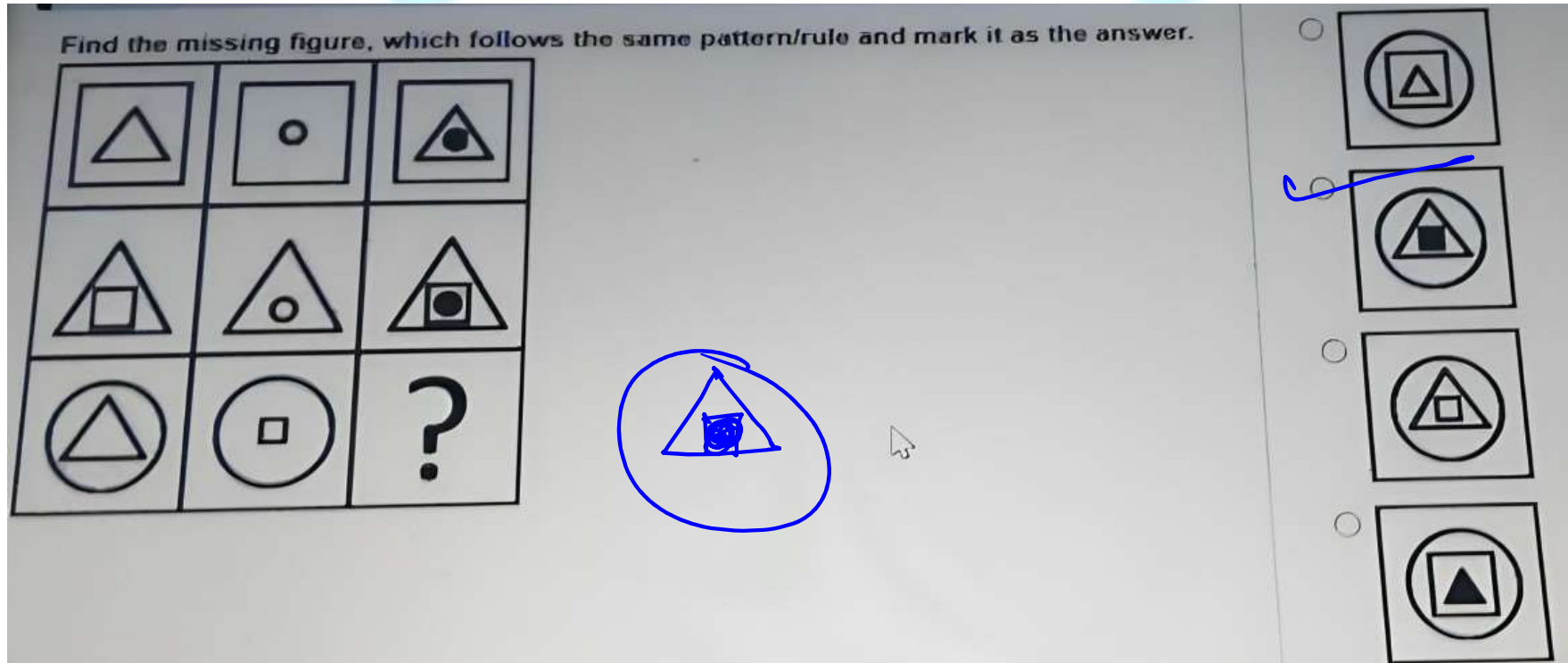


Susan,



Robert

31.



32.

Find the missing figure, which follows the same pattern/rule and mark it as the answer.

		?

Handwritten notes and options:

a b c

$a + c - (a \cap c)$

Options (marked with radio buttons):

- ☒
- ☐
- ☐
- ☐



33. Consider the following statements and determine which of the given conclusions follow.

Statements:

a. All bats are phones.

b. All lamps are phones.

Conclusions:

I. Some lamps are bats.

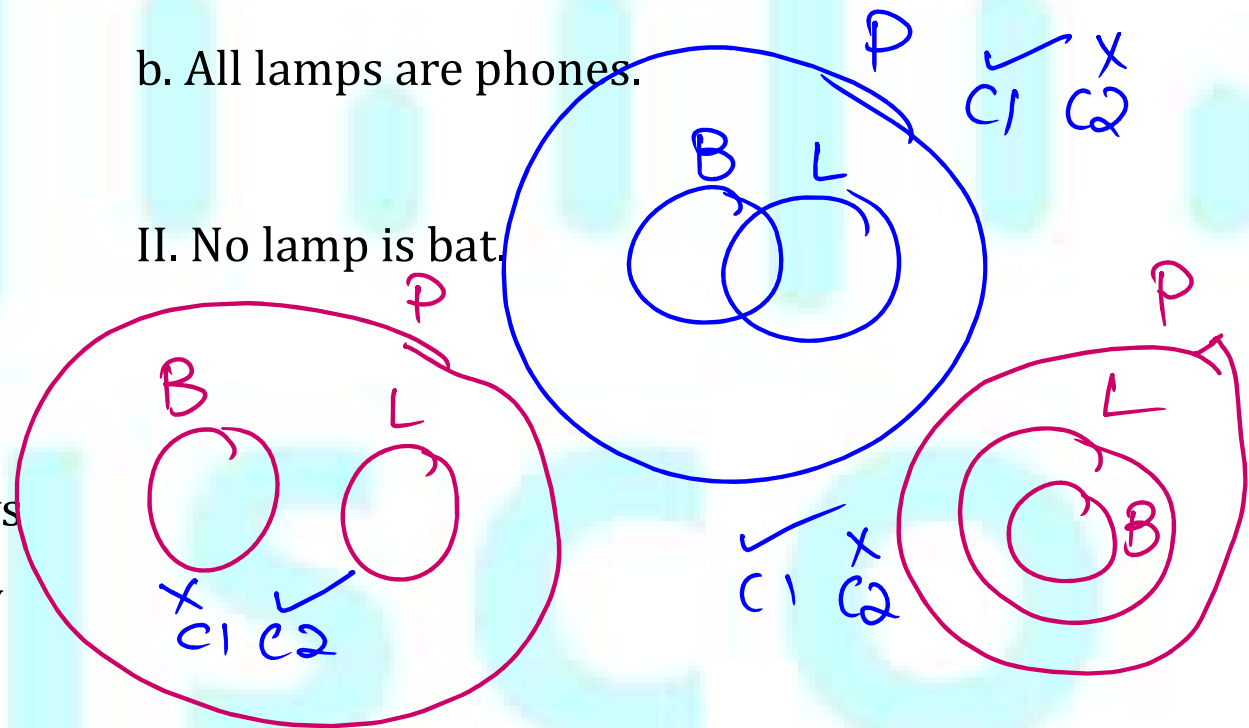
II. No lamp is bat.

A. Only I follows

B. Only II follows

✓ C. Either I or II follows

D. Both I and II follow





34. John is taller than Kris. Leonard is taller than Mario.

Mario is shorter than Nick. Kris and Nick have exactly the same height.

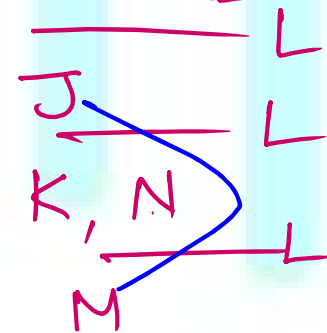
If the information above is true, which of the following must also be true?

A. Leonard is taller than Nick.

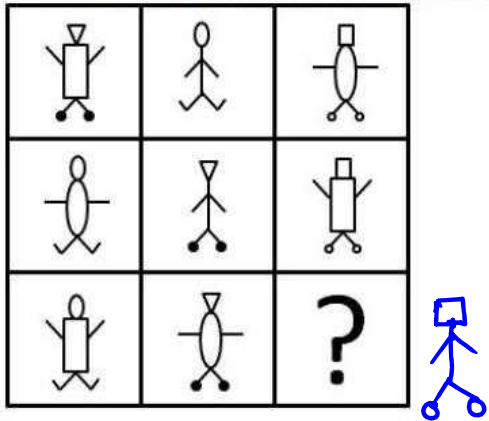
B. Leonard is taller than John.

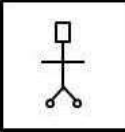
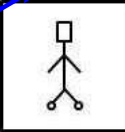
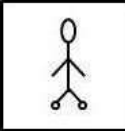
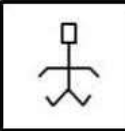
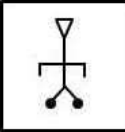
C. Kris is shorter than Leonard.

☒ D. John is taller than Mario.



Find the missing figure, which follows the same pattern/rule and mark it as the answer.



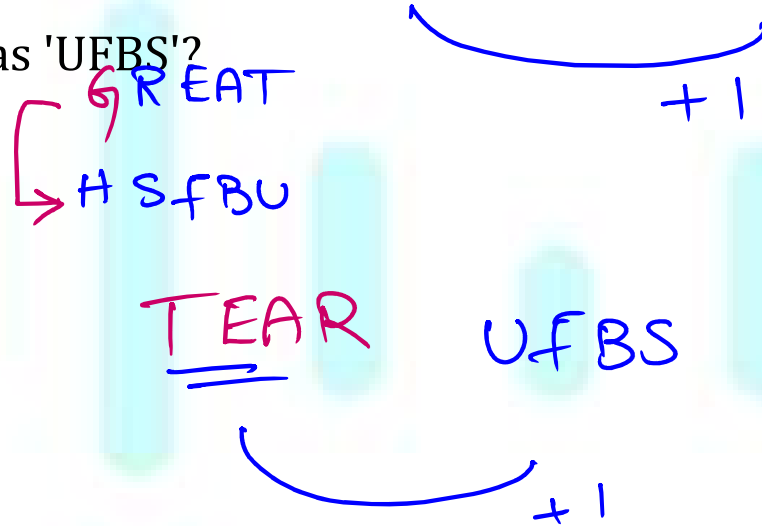
- ☐ 
- ☒ 
- ☐ 
- ☐ 
- ☐ 

35.



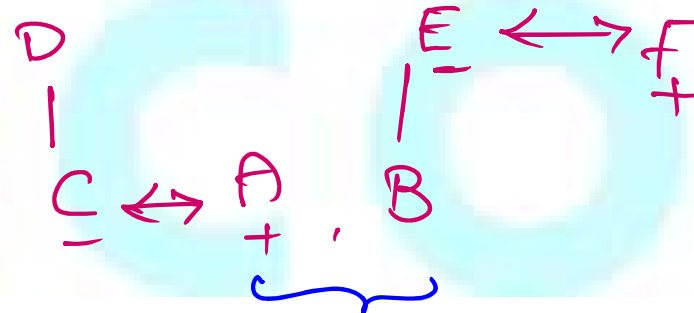
36. In a certain coding system 'GREAT' is coded as 'HSFBU'. Which of the following word is coded as 'UFBS'?

- ☒ A. TEAR
- B. VGCT
- C. VCGT
- D. TERA



37. A is B's brother, C is D's daughter, C is wife of A, E is mother of B, and F is father-in-law of C. How is E related to F?

- A. Sister
- ☒ B. Wife
- C. Daughter-in-law





D.Brother

38. In a certain code language GOVERNOR is written as OWDMZVWZ. How is PENUMBRA written in that code language?

~~A. XMWCUJZI~~

~~B. XMVCUJZK~~

~~C. XMVDTJZI~~

~~D. XMVCUJZI~~

GOVERNOR

OWDMZVWZ

+8

$V + 8 = 29 \rightarrow C$

PENUMBRA

1 1 1 1 1 1
X M V U J Z I
9



$$P = 20$$

$$Q = 10 \text{ (assume)}$$

39. 1. P sold is double the number of cookies sold by Q.

2. The number of cookies sold by R is 50% of the number of cookies sold by Q.

3. P sold is four-times the number of cookies sold by R.

$$R = \frac{1}{2} Q = 5$$

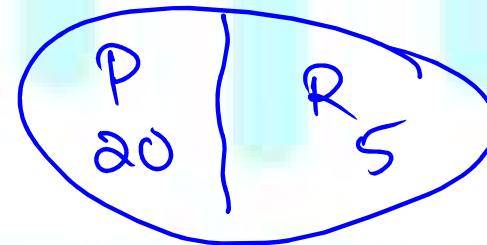
If the first two statements are true, the third statement is:

✓ A. True

B. False

C. Uncertain

$$P = 4R$$



40. Consider the following statements and determine which of the given conclusions follow.



Statements:

A. No man is a donkey.

B. Raj is a man. — All Raj are men

Conclusions:

I. Raj is not a donkey.

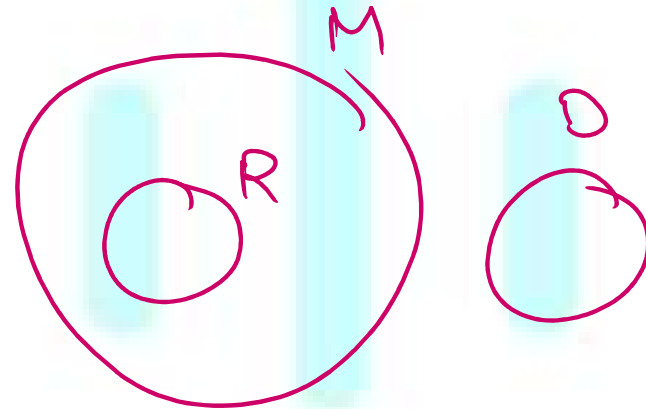
II. All men are not Raj. — No man is Raj

✓ A. Only conclusion I follows

B. Only conclusion II follows

C. Either conclusion I or II follows

D. Neither conclusions I or II follow



C1 ✓ C2 X



41. Five friends, A, B, C, D, and E, with different occupational backgrounds (Teacher, Doctor, Accountant, Businessman, and Engineer) were sitting in a row. The doctor sat next to B. The engineer sat between B and the accountant. A and the accountant were sitting next to each other but none of them sat at the extreme ends. What is the occupation of A?

A. Teacher

~~B. Engineer~~

C. Doctor

D. Businessman

— — — A B —
 ← →
 Acc Engg Doctor

— B A — —
 ↑
 Doctor Engg Ac



42. Two outlet pipes can empty a tank containing 1300 litre of water in 7.2 minutes.

What is the rate at which the tank gets emptied?

- ☒ A. 325 L/min
- ☒ B. 394 L/min
- ☒ C. 180.56 L/min
- ☐ D. 135.44 L/min
- ☒ E. 481.5 L/min

Pipes & Cisterns

$$\frac{1300 \text{ lit}}{7.2 \text{ min}} \approx 180.56$$



43.

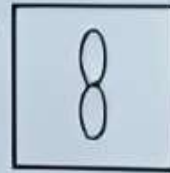


Find the missing figure, which follows the same pattern/rule and mark it as the answer.

		?

$C = A \cap B$



- ☐ 
- ☐ 
- ☐ 
- ☐ 
- ☒ 



44. Consider the following statements and determine which of the given conclusions follow.

Statements

a. Some dentists are dumb

b. Some dumb are rotten

Conclusions

I. Some dentists are rotten.

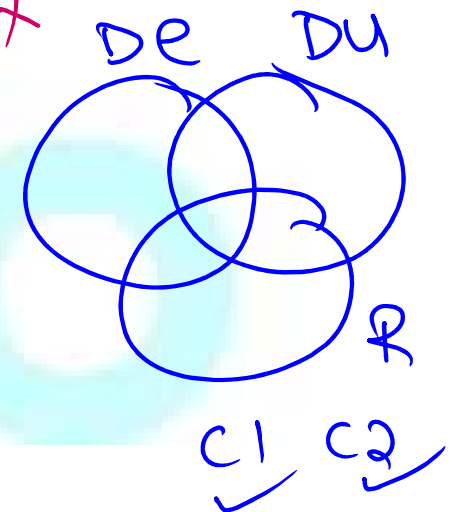
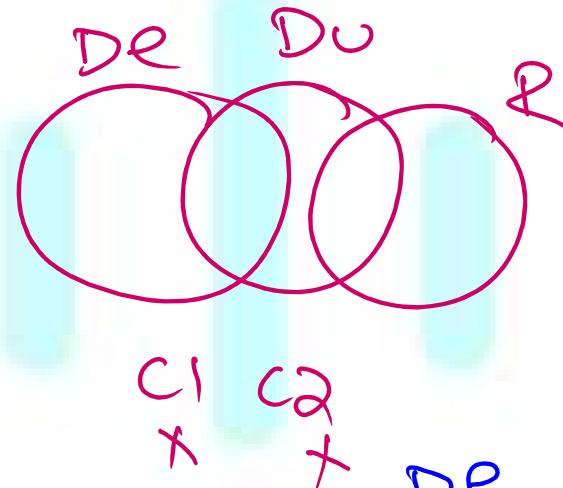
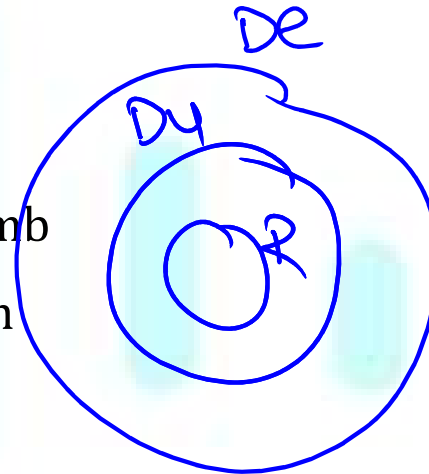
II. Some rotten are dentists

A. Only I follows

B. Only II follows

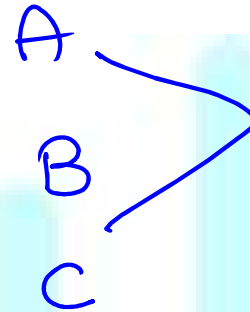
C. Either or II follows

☒ D. Neither follow





45. 1. Pool A is deeper than pool B
2. Pool B is ~~shallower~~ ^{deeper} than pool C
3. Pool C is deeper than pool A



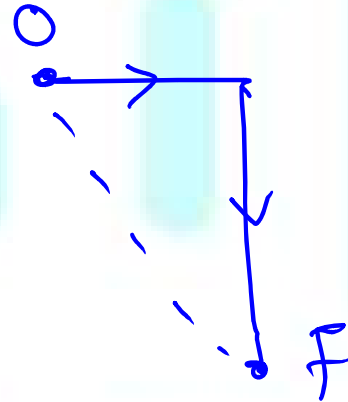
If the first two statements are true, the third statement is

- A. True
- B. False ✓
- C. Uncertain



46. Alan starts from his office and runs 5 miles in the East direction. He then turns right and runs 10 miles more. In which direction is he with respect to his office?

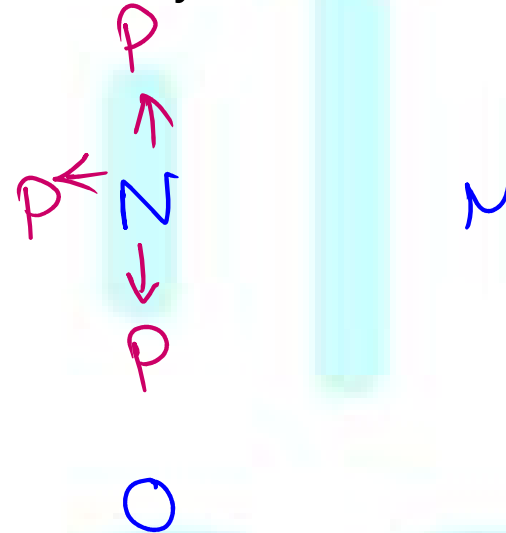
- ✓ A. South-East
- B. North
- C. West
- D. North-West





47. City 'M' is situated east of city 'N'. City 'O' is situated south of city 'N'. City 'P' is situated north of city O. In which direction is the city 'N' situated with respect to city 'P'?

- A. North
- B. South
- C. East
- ✓ D. Cannot be determined



48. 1. Sam scored more marks in English than Alex.
2. Alex scored lesser marks in English than Carol.
3. Carol doesn't score highest in English among Sam, Alex and herself.
If the first two statements are true, the third statement is:



A. True

B. False

☒ C. Uncertain

Repeated

49. Consider the following statements and determine which of the given conclusions follow.

Statements:

a. All bats are phones.

b. All lamps are phones.

Conclusions:

I. Some lamps are bats. II. No lamp is bat.

A. Only I follows

Repeated



B. Only II follows

✓ C. Either I or II follows

D. Both I and II follow

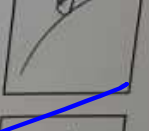
50.

Find the missing figure, which follows the same pattern/rule and mark it as the answer.

		
a	+b	=c
		
		

Revisit

Choose the best option

- ☐ 
- ☐ 
- ☐ 
- ☐ 
- ☒ 
- ☐ 